



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

STATISTICS
PAPER 1

6046/1
3 hours

SPECIMEN PAPER

Additional materials:
Answer booklet
Graph paper
List of Formulae MF 7
Non programmable Scientific calculator
Mathematical Instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.

Answer **all** questions.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 120.

This specimen paper consists of 6 printed pages.

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Answer **all** questions

1. (a) State any **two** advantages of a stem and leaf diagram. [2]
- (b) Twenty students sat for a Statistics test and their marks were recorded as follows:
- 33; 30; 31; 32; 29; 43; 47; 45; 38; 28; 34; 41; 19;
17; 50; 41; 19; 17; 50; 20.
- (i) Draw a stem and leaf diagram to illustrate the data. [4]
- (ii) Comment on the nature of the distribution. [1]

2. The discrete random variable Y has the probability distribution shown below:

y	1	2	3	4	5
$P(Y=y)$	0.06	0.09	b	$2b$	0.06

Given that b is a constant, find the

- (a) value of b . [3]
- (b) expectation of y . [2]
- (a) A bag contains ten identical pens of which four are red and six are blue. An experiment consists of drawing at random, three pens from the bag one at a time without replacement.
- Draw a tree diagram to show all the possible outcome of the experiment. [5]
- (b) Hence or otherwise, find the probability that all three pens are of the same colour. [3]

4. (a) Define the following terms:

(i) *permutation*,

[1]

(ii) *combination*,

[1]

(b) Nine different books are to be arranged in empty spaces on a bookshelf.
Find the number of ways in which the nine different books can be arranged.

[2]

(c) Find the number of ways of selecting 11 football players from a team of 15.

[2]

5. (a) Bhekani scored the following number of goals in eleven matches.

1: 0: 0: 2: 2: 0: 1: 2: 3: 1: 2.

Write down the modal number of goals.

[1]

(b) The marks scored by 21 students in a Mathematics test are shown in the table below:

Mark	0	1	2	3	4	5
Frequency	2	7	4	1	5	2

(i) Find the median.

[1]

(ii) Calculate the mean mark, giving the answer to three significant figures.

[2]

6. A hotel kitchen has two food mixers A and B. The number of times per week that A and B break down follow a Poisson distribution with means 0.4 and 0.1 respectively.

Find to three decimal places, the probability that in the next three weeks,

(a) A will not break down,

[2]

(b) there will be a total of two breakdowns,

[2]

(c) there will be at least three break downs.

[3]

7. A random sample of 75 observations from a normal population with mean μ gave the following summarised data:

$$\begin{aligned}\sum (x - 80) &= -60 \\ \sum (x - 80)^2 &= 2110\end{aligned}$$

Calculate the

- (a) mean, [2]
- (b) unbiased estimate of the population variance and standard deviation, [4]
- (c) 95% confidence interval for the mean. [3]

8. The heights of students at a certain school in Mashonaland West are normally distributed with a mean height of 150 cm and a standard deviation of 20 cm.

If a student is chosen at random calculate the probability that the student's height is

- (a) greater than 160 cm, [4]
- (b) between 160 cm and 180 cm. [4]

9. The continuous random variable x has a probability density function $f(x)$ where

$$f(x) = \begin{cases} c(x+2)^2 & -2 \leq x < 0 \\ 4c & 0 \leq x \leq \frac{4}{3} \\ 0 & \text{otherwise} \end{cases} \text{ where } c \text{ is a constant}$$

- (a) Find the value of c . [3]
- (b) Sketch the graph of $y = f(x)$. [3]
- (c) Find $P(-1 \leq x \leq 1)$. [4]
- (d) Find $P(x > 1)$. [2]

10. (a) List any **two** features of a Normal Distribution. [2]
- (b) The dimensions of metal strips produced by a machine are normally distributed with a mean length of 145 cm and a standard deviation of 10 cm.
Find the probability that the randomly selected strip is
- (i) shorter than 160 cm, [3]
- (ii) within 5 cm of the mean. [4]
11. A random sample of 100 observations from a normal population with mean μ gave the following data.
 $\Sigma x = 8\,200$, $\Sigma x^2 = 686\,800$
 Find a 98% confidence interval for μ . [8]
12. (a) State any **two** conditions necessary for a random variable to be modelled using a Poisson Distribution. [2]
- (b) $X \sim Po(4.5)$
 Find the
- (i) mean, [1]
- (ii) standard deviation. [2]
- (iii) $P(X \geq 6)$ [4]

13. (a) State any **two** methods that can be used to smoothen a time series. [2]
- (b) State any **two** advantages of smoothening a time series. [2]
- (c) The table below shows the marks scored by 12 learners in a Mechanics test.

Learner	Score
A	10
B	55
C	46
D	27
E	17
F	12
G	85
H	64
I	36
J	40
K	30
L	37

Using the information on the table above to

- (i) draw a time series graph of the Mechanics test score, [5]
- (ii) calculate the 3-point moving average, giving the answer to 3 significant figures, [4]
- (iii) plot the 3 point moving averages scores on the same axes as of the time series graph in (i) above. [3]

4. The number of accidents per day on a stretch of Masvingo Harare highway was recorded for 100 days and the following results were obtained.

Number of accidents	0	1	2	3	4	5 or more
Frequency	44	32	9	10	5	0

- (a) Calculate the mean number of accidents per day. [2]
- (b) Test at 5% level of significance whether Poisson model is suitable to represent the number of accidents per day. [14]

